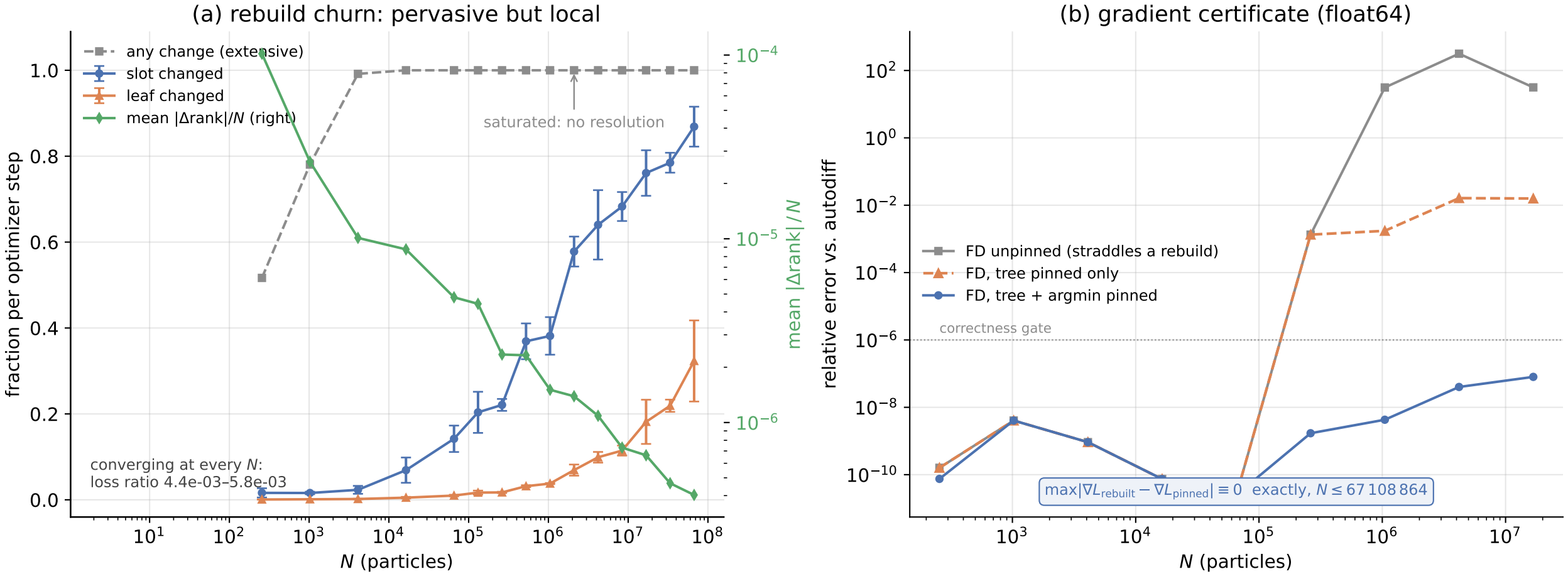
(a) rebuild churn: pervasive but local
any change (extensive)
slot changed
leaf changed
mean |Δrank|/N (right)
saturated: no resolution
converging at every N:
loss ratio 4.4e-03–5.8e-03
fraction per optimizer step
mean |Δrank| / N
N (particles)
(b) gradient certificate (float64)
FD unpinned (straddles a rebuild)
FD, tree pinned only
FD, tree + argmin pinned
correctness gate
max|∇L_rebuilt − ∇L_pinned| ≡ 0 exactly, N ≤ 67 108 864
relative error vs. autodiff
N (particles)